

What will this expression create in Python? `x=[1, 2.3, 4.5, 1.4]`

0 0 0 0 0 0

A tuple A dictionary An array A vector A function A list

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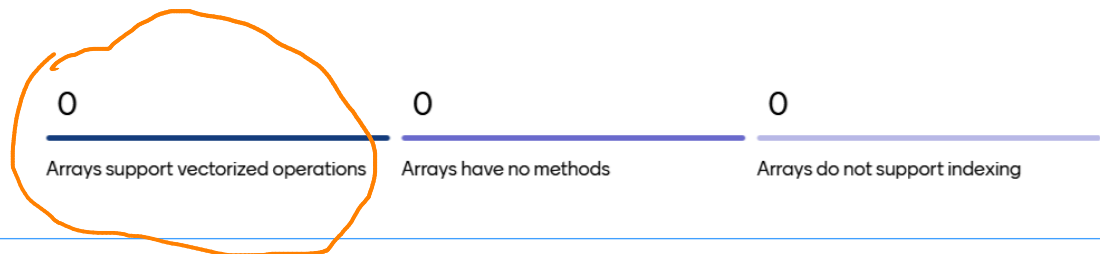
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When might you use a tuple instead of a list?

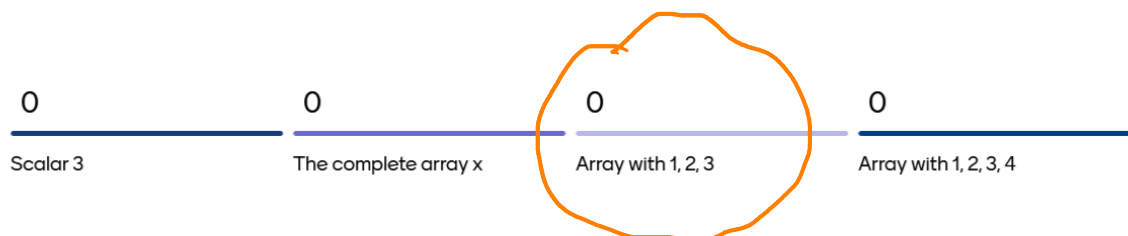
0 0 0

For model parameters For 2D data For vectorized operations

How do NumPy arrays differ from lists?



If you run `x = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10])` and then `x[:3]`, what will you get?



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What is an easy way to create an array that consists of 100 zeros?

All responses to your question will be shown here

Each response can be up to 200 characters long

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```
import numpy as np  
np.zeros(100)
```

A hot (393 K) surface (0.1 m²) is cooling in air with $h = 150 \text{ W/m}^2\cdot\text{K}$. How quickly does it lose heat?

What is the heat flux in kW/m²?

15 kW/m²

What is the heat flow in kW?

1.5 kW

0

10 50

$$q = h \cdot (393 \text{ K} - 293 \text{ K})$$

$$= 150 \cdot 100 = 15000 \text{ W/m}^2 = 15 \text{ kW/m}^2$$

It took 100 J of heat to completely melt some ice. How much is then needed to increase its temperature from 0 to 10 deg. C?

$$100 \cdot 0.12 = (12) \text{ J}$$

How much (in J)?

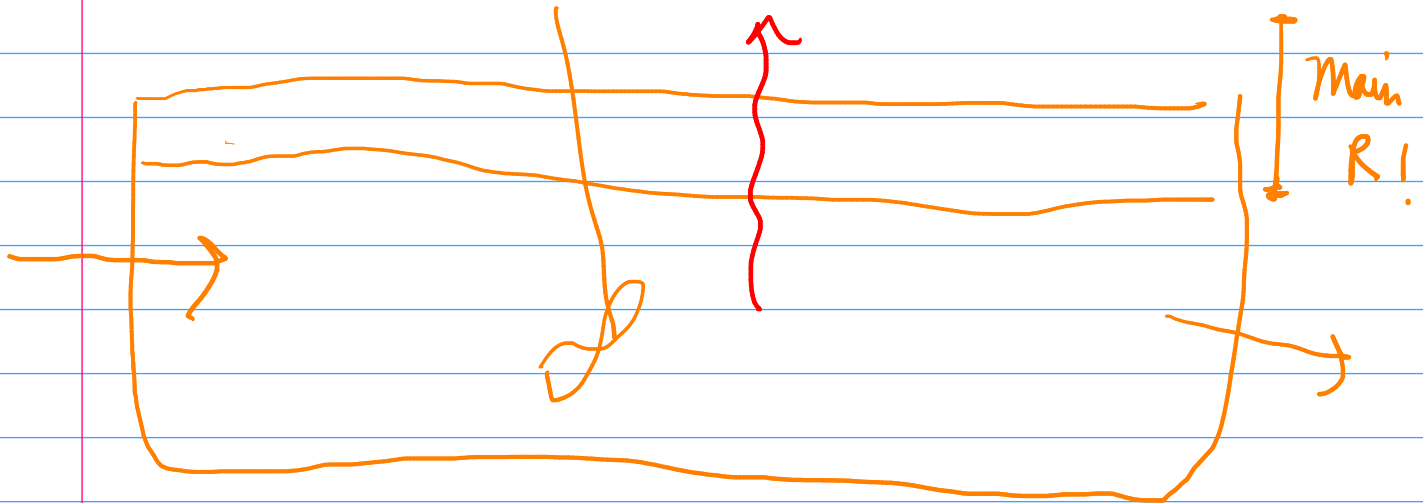
or ~ 10

0

100

$$\frac{\text{specific heat} \cdot 10^\circ\text{C}}{\text{heat of fusion}} = \frac{4.2 \text{ J/g}\cdot\text{K} \cdot 10 \text{ K}}{333 \text{ J/g}} = 0.12 \text{ J/g}$$

A mixed and heated swimming pool has an insulated cover on top. What approach might you use for a heat loss model and why?



I would try to use the lumped capacitance approach using an overall heat transfer coefficient U that includes the insulated cover and a convection heat transfer coefficient above.

This should work because most of the resistance must be in the insulated cover and air film above the cover.

Why? Well, 1) water has higher thermal conductivity than air anyway, 2) mixing reduces resistance further, and 3) the insulation adds more resistance at the interface.

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The wall of your flat has several layers. What approach could you use to model average heat transfer through it?

$$U \text{ or } R = \frac{1}{U}$$

